

Metallic and non-metallic continuum

Learning outcomes

By the end of this section you should be able to:

- Describe the trend from basic metal oxides through amphoteric to acidic non-metal oxides.
- Deduce equations for the reactions with water of the oxides of group 1 and group 2 metals, carbon and sulfur.

You will notice in chemistry that many of the categories used to label substances actually show a continuum across the categories. We saw this in [section S2.4.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/>) as metallic and non-metallic elements showed a continuum in their bonding. Consider the elements in period 3 (Na, Mg, Al, Si, P, S, Cl). How would you classify the type of each of these elements? How would you classify the bonding each of these elements have?

Compounds of metallic and non-metallic elements also show a continuum in their bonding. Consider the bonding triangle for the oxides of period 3 elements (Na_2O , MgO , Al_2O_3 , SiO_2 , P_4O_{10} , SO_2/SO_3 , Cl_2O), what type of bonding do you expect each oxide to have? Compare your expectations to the bonding triangle in **Figure 1**.

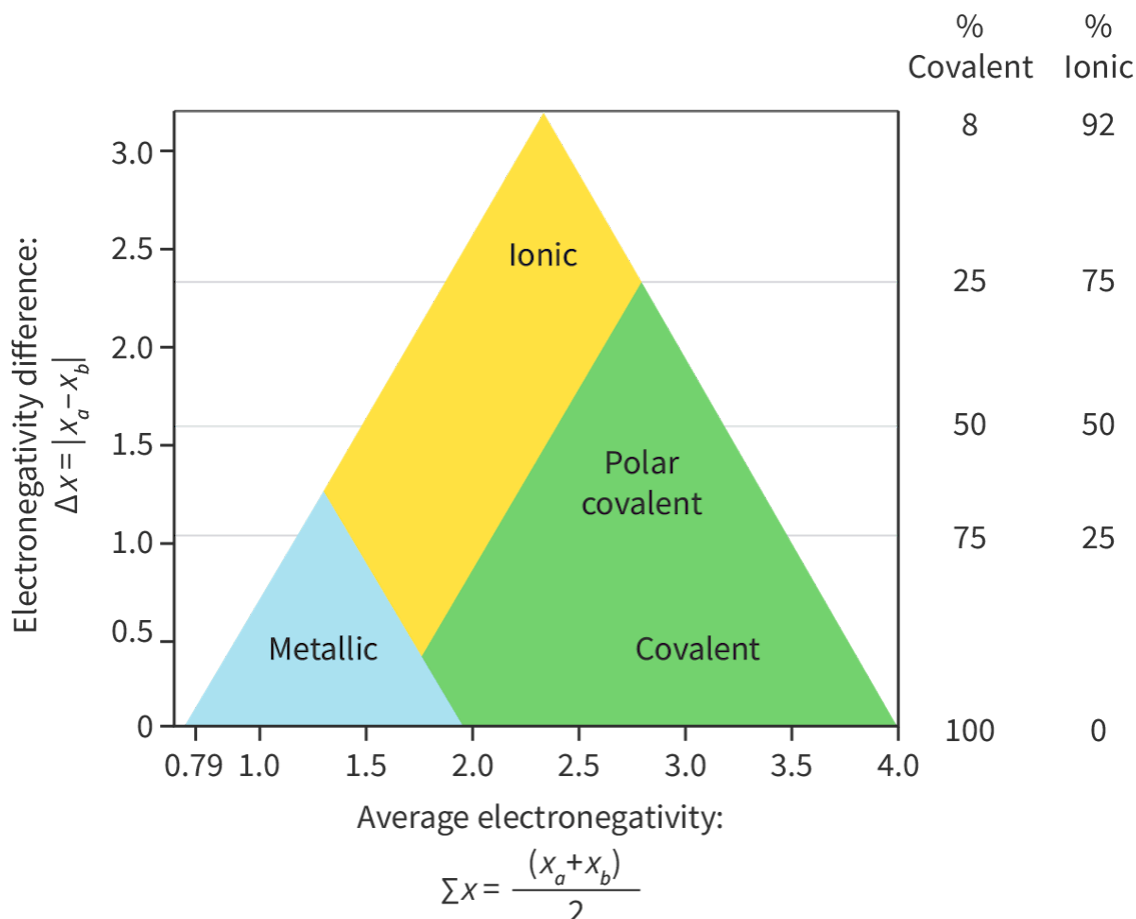


Figure 1. Continuum of bonding of the period 3 oxides.

 More information for figure 1

The image is a triangle diagram that categorizes chemical bonds according to average electronegativity and electronegativity difference. The diagram is divided into three sections: metallic, polar covalent, and ionic.

- The X-axis represents average electronegativity ranging from 0.79 to 4.0. The formula ($\Sigma x = \frac{(x_a + x_b)}{2}$) is shown on this axis.
- The Y-axis shows electronegativity difference, ranging from 0 to 3.0, noted as ($\Delta x = |x_a - x_b|$).
- The bottom left triangle is labeled "Metallic," covering lower values on both axes. This area suggests bonds with low average electronegativity and small differences.
- The upper triangle is labeled "Ionic," indicating high electronegativity differences, regardless of average values.
- The middle section is labeled "Polar covalent," bridging the gap between metallic and ionic areas.

There is also additional information on the right indicating the percentage composition of covalent vs. ionic character as a gradient, with extremes marked at the top as 8% covalent and 92% ionic, respectively.

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What do you expect about the structure and properties of metallic and non-metallic elements and compounds? The structure of metallic and non-metallic elements and compounds also show a continuum. The structure of the elements and oxides of period 3 are shown in **Table 1**.

Table 1. Structure of the elements and oxides of period 3

Element s & Their Structure	Element s	Na	Mg	Al	Si	P
	Structure	Metallic lattice	Metallic lattice	Metallic lattice	Covalent network lattice	Covalent molecule (P ₄)
Oxides & Their Structure	Oxides	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀
	Structure	Ionic lattice	Ionic lattice	Ionic lattice	Covalent network lattice	Small cage- like structure

Acid-base nature of metallic and non-metallic oxides

Another property that shows a continuum is the acid-base nature of metallic and non-metallic oxides. Can you think of any acids you have encountered before in your personal experience? Lemons and vinegar are sour and acidic because of the

acid molecules present within them that form H^+ ions in solution. Cleaning products and soaps are slippery and basic because of the substances within them that can form OH^- ions in solution. Note that this section is considering acids and bases more generally. You will look at them in more detail in [subtopic 3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>).

A quantitative measurement to determine the concentration of H^+ ions in solution is called the pH scale. Acids are classified as solutions with a pH between 0 and 7 where bases are classified as solutions with a pH between 7 and 14. **Figure 2** shows common substances you may have encountered and where they fall on the pH scale. You will look more closely at the pH scale in [section R3.1.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/) ([https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-\[h+\]-id-44702/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/)).

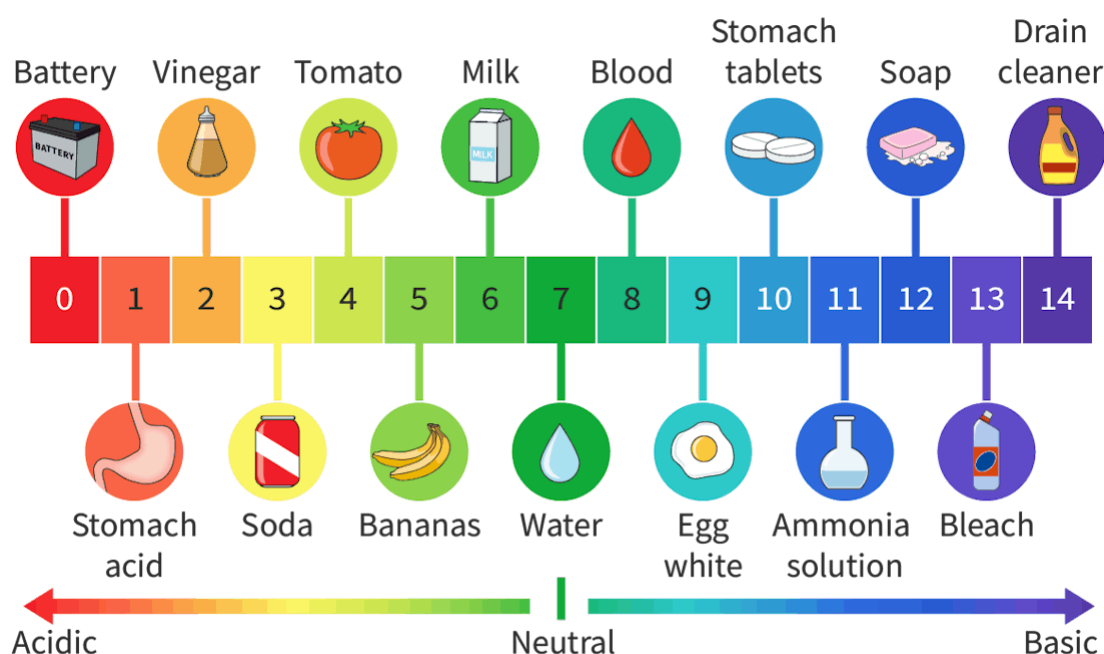


Figure 2. pH of common substances.

 More information for figure 2

This diagram illustrates the pH scale, which ranges from 0 to 14, with various common substances placed along the scale according to their acidity or basicity. The scale begins with 0, marked with a battery icon, indicating highly acidic. Moving along the scale to 7, it shows neutral substances like water. Substances are depicted with images: battery (pH 0), stomach acid (pH 1), vinegar (pH 2), tomato (pH 4), milk (pH 6), blood (pH 7), stomach tablets (pH 10), and soap (pH 12), among others, culminating in drain cleaner at pH 14, which is highly basic. An arrow runs beneath the scale, pointing from acidic to basic, with a central point labeled "Neutral."

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Period 3 oxides show a continuum in their acid base nature. **Table 2** shows the acid base character of the period 3 oxides. You can see the continuum from basic metal oxides to acidic non-metal oxides. You might notice that Al_2O_3 is listed as **amphoteric**. The term amphoteric means that a substance has the ability to act as either an acid or a base depending on the surrounding environment. Notice in **Figure 1** how Al_2O_3 also falls right at the ionic/covalent cusp of the bonding triangle providing further evidence for why it can sometimes act like a basic metal oxide and sometimes act like an acidic non-metal oxide.

Table 2. Acid-base character of the period 3 oxide

Oxides	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}
Acid-base character	Basic		Amphoteric		
pH	High (>7)		Powder (>7)) in water (<7)		

Chemical reactions for oxides

We can look at some of the reactions of period 3 oxides with water to identify the acids and bases produced in **Figure 3** and **Figure 4**.

Basic oxides

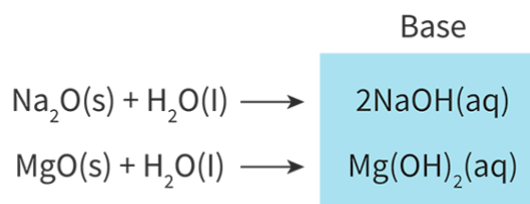


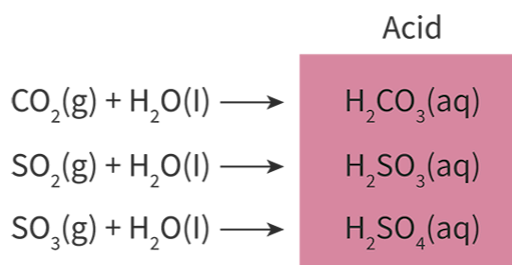
Figure 3. Reactions of metal oxides with water forming a strong base. More information for figure 3

The diagram illustrates the reactions between metal oxides and water, resulting in the formation of metal hydroxides. Two reactions are depicted with chemical formulas and directional arrows indicating the relationship between reactants and products. The first reaction shows a metal oxide reacting with water to form $2\text{NaOH}(\text{aq})$, suggesting the involvement of sodium. The second reaction similarly depicts an oxide reacting with water to produce $\text{Mg}(\text{OH})_2(\text{aq})$, indicating the formation of magnesium hydroxide. The diagram highlights that these reactions lead to the formation of strong bases, as metal hydroxides are released in aqueous form. This visual representation underscores the chemical transformation of metal oxides into basic solutions upon interaction with water.

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When metal oxides react with water, a metal hydroxide is formed. Metal hydroxides are ionic compounds and the chemical formula of the metal hydroxide formed can be deduced by considering the charge on the metal ion and the 1– charge on the hydroxide ion as you learned in [section S2.1.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>). The metal hydroxide formed is a base leading to the high pH of aqueous solutions containing metal oxides. You will learn more about bases in [section R3.1.6](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>).

Acidic oxides

**Figure 4. Reactions of non-metal oxides with water forming an acid.** More information for figure 4

The diagram illustrates the reactions of various non-metal oxides with water resulting in the formation of acids. Starting from the top, carbon dioxide (CO_2) is shown reacting with water (H_2O) to produce carbonic acid (H_2CO_3). Next, sulfur dioxide (SO_2)

combines with water to form sulfurous acid (H_2SO_3). Finally, sulfur trioxide (SO_3) reacts with water, giving rise to sulfuric acid (H_2SO_4). Each reaction is depicted with an arrow pointing from the non-metal oxide and water towards the resulting acid. The chemical formulas are clearly displayed next to each reaction arrow.

[Generated by AI]

Non-metal oxides react with water to produce acids. For example, carbonic acid, H_2CO_3 , is produced when carbon dioxide reacts with water. And when sulfur dioxide or sulfur trioxide react with water, sulfurous acid, H_2SO_3 or sulfuric acid, H_2SO_4 are produced. You will learn more about acids in [section R3.1.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>).

Study skills

Note: you do need to be able to deduce the equations for the reactions with water of the oxides of group 1 and group 2 metals, carbon and sulfur. Most should be fairly straightforward as the products formed are examples of ionic formulas using polyatomic ions you encountered in [section S2.1.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>). SO_2 is the less intuitive example and worth spending additional time reviewing to familiarise yourself with it.

Implications of acidic non-metal oxides

The ability of non-metal oxides to form acids when they react with water has some major implications for our ecosystems, our health and our infrastructure. The two major global concerns of these acidic non-metal oxides are:

1. ocean acidification
2. acid rain.

Where might you have heard of these terms before? What sort of impact do you expect these to have on ecosystems?

Ocean acidification

Did you notice in **Figure 4** how when CO_2 interacts with water an acid, H_2CO_3 is formed? Can you think about why this might be a global concern? How is CO_2 impacting our earth's atmosphere? What are some major sources of CO_2 ?

The accumulation of CO_2 in the Earth's atmosphere, primarily due to the combustion of fossil fuels, has major implications for our environment as shown in **Figure 5**. Not only does CO_2 act as a greenhouse gas, but it also can dissolve in bodies of water, acidifying it as H_2CO_3 is formed. This has major implications for our oceans in particular as marine organisms have difficulty adapting to minor pH changes of their environment. Ocean acidification can affect societies as marine food chains and storm protection from reefs are negatively impacted.

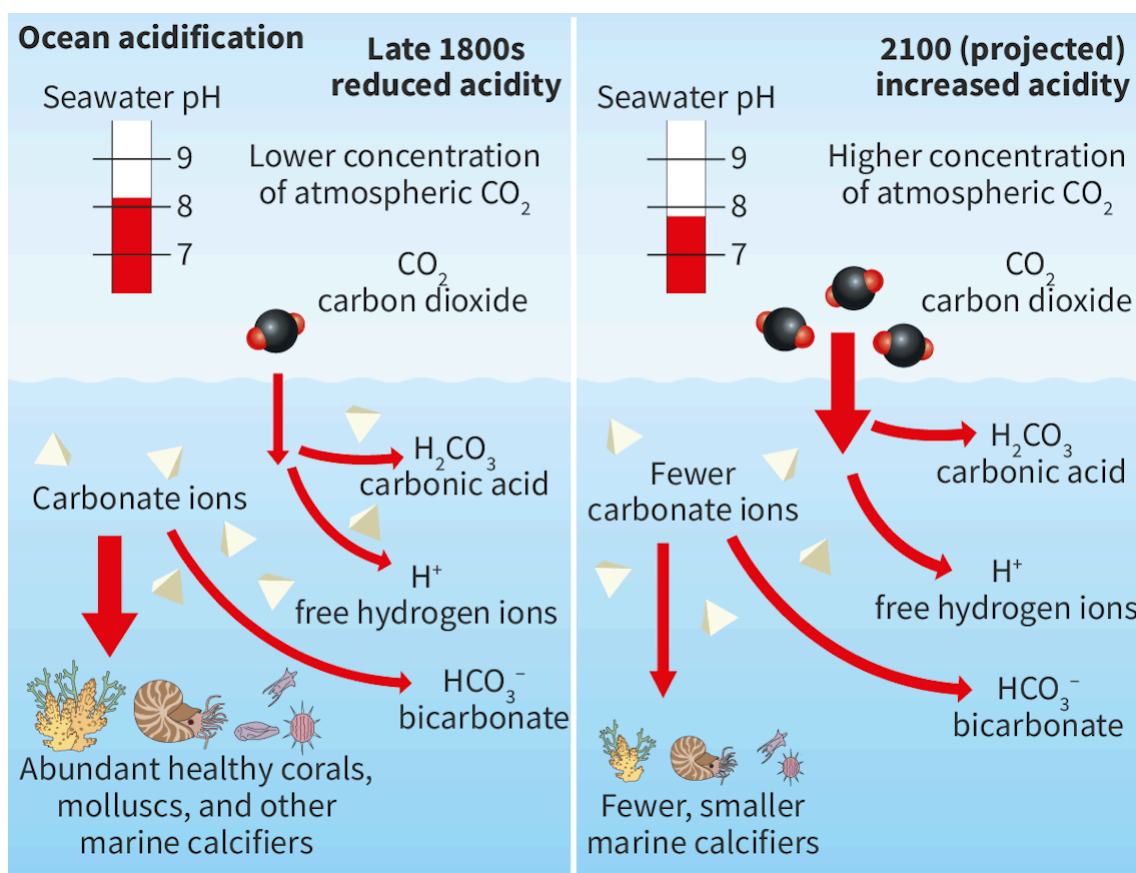


Figure 5. Effects of ocean acidification on marine life.

 More information for figure 5

The diagram illustrates the effects of ocean acidification on marine life comparing two time periods: the late 1800s, representing reduced acidity, and the projected 2100, showing increased acidity. On the left, the late 1800s section shows a lower concentration of atmospheric CO_2 . Carbon dioxide dissolves in seawater, forming carbonic acid (H_2CO_3), which then dissociates into free hydrogen ions (H^+) and carbonate ions. This environment supports abundant, healthy corals, mollusks, and other marine calcifiers. The pH scale on the left indicates seawater pH ranging from 9 to 7, where lower acidity is indicated by a higher pH.

On the right, the 2100 (projected) increased acidity section depicts a higher concentration of atmospheric CO₂. More carbon dioxide leads to more carbonic acid formation, resulting in more free hydrogen ions, thereby reducing carbonate ions available to marine organisms. Consequently, fewer and smaller marine calcifiers, such as corals and mollusks, are evident. The pH scale also ranges from 9 to 7, but this time illustrates increased acidity as represented by a lower pH. Arrows in the diagram trace the transformation of CO₂ to carbonic acid and further to bicarbonate, highlighting the decreasing carbonate availability in the future scenario.

[Generated by AI]

Compare the two photos in **Interactive 1** showing thriving healthy coral and coral bleached from rising temperatures and acidified oceans.

Interactive 1. Comparison of Healthy and Bleached Coral.

 More information for interactive 1

The interactive has two slides that display coral reefs grown under two different environmental conditions, showing the impact of climate change on marine ecosystems.

Slide 1: In this interactive, the coral reef is healthy. The corals are brightly colored in shades of orange, pink, yellow, and green, indicating that they are alive and covered with algae. These algae provide food to the corals through photosynthesis, giving the reef its vibrant colors. A variety of marine species depend on this coral reef for shelter and survival. The water appears clear, suggesting a balanced and stable ecosystem. The text below the image reads, thriving healthy coral.

Slide 2: In this interactive, the coral reef is completely bleached. The colorful corals

have turned white. This phenomenon, known as coral bleaching. The text below the image reads, bleached coral caused by rising temperature and acidification of the ocean.

Comparing the two interactives highlights the devastating impact of climate change and environmental stressors on coral reefs. Coral bleaching is a warning sign of ocean health deterioration, emphasizing the need for conservation efforts to protect marine life and mitigate climate change effects.

Acid rain

Have you heard of acid rain before? What do you think it means? Where do you think the acid in acid rain comes from? Two gaseous non-metal oxides are the major contributors to acid rain. Sulfur and nitrogen containing oxides readily produce the acids H_2SO_4 and HNO_3 , when they react with water. Any form of deposition (rain, sleet, snow, fog) that contains H_2SO_4 and/or HNO_3 is referred to as acid rain. Since H_2SO_4 and HNO_3 form easily when gaseous oxides of sulfur and nitrogen interact with water, it is events producing these gases that are the root causes of acid rain.

Sulfur oxides can be produced naturally from volcanoes and nitrogen containing oxides can be produced naturally in a thunderstorm, but these quantities are small to negligible in comparison to the volumes produced by industrial activity. Sulfur is an impurity in fossil fuels extracted from the ground. When these are burned SO_2 (which easily reacts with O_2 in the air to form SO_3) are released into the atmosphere. Nitrogen oxides are produced when NO (which easily reacts to form NO_2) is formed when N_2 in the air reacts with oxygen during a combustion reaction of fossil fuels.

The formation of acid rain has significant environmental concerns on both land and aquatic ecosystems. These range from impacting crop yields, to damaging infrastructure, impacting tree growth and the negatively impacting the respiratory systems of mammals. These are illustrated in **Figure 6**.

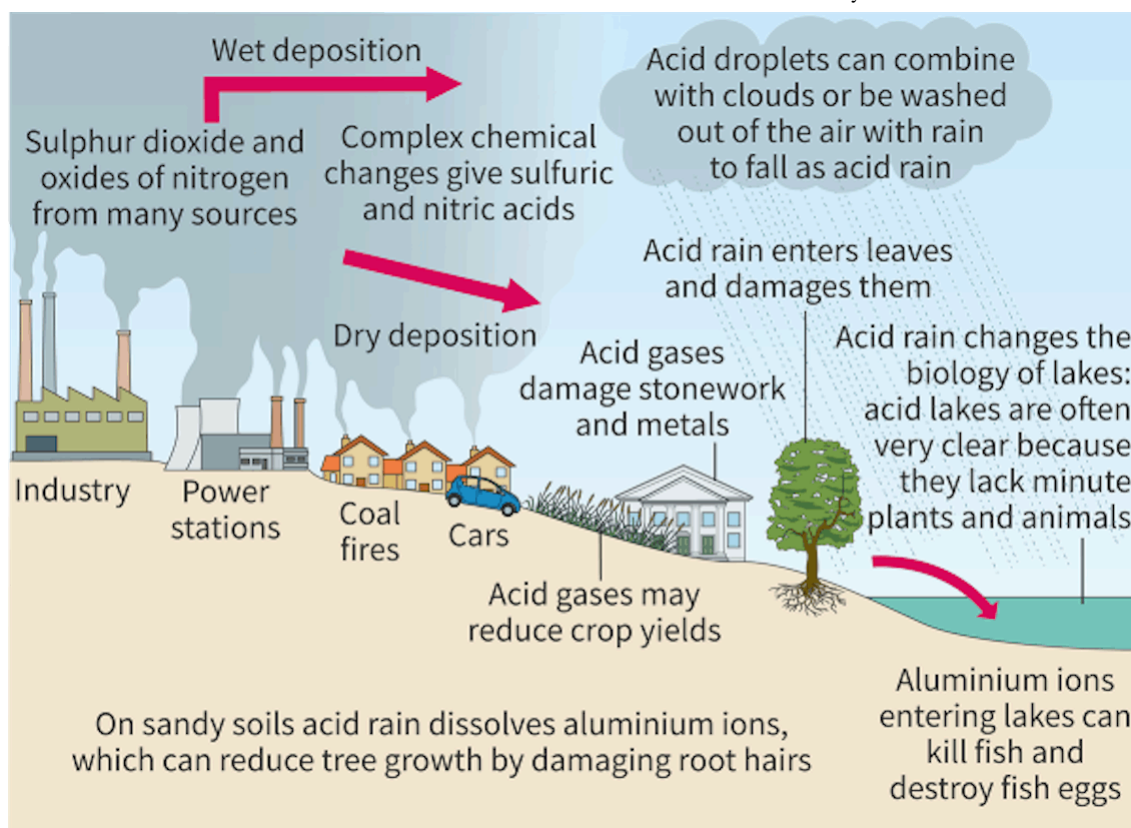


Figure 6. Causes and effects of acid rain.

 More information for figure 6

The diagram presents the causes and effects of acid rain with various labeled elements and arrows indicating the flow of causes to effects. The left side shows industrial factories, power stations, coal fires, and cars emitting sulfur dioxide and nitrogen oxides, leading to complex chemical changes that produce sulfuric and nitric acids. These emissions are transported as wet or dry deposition. Arrows labeled "wet deposition" and "dry deposition" point toward the right, indicating how these acids fall to the ground as acid rain.

On the right side, the effects of acid rain are shown. Arrows indicate that acid rain can enter leaves and damage them, affect the biology of lakes by making them clear due to lack of organisms, and harm stonework and metals. Text specifies that acid gases may reduce crop yields and notes that acid rain's effects on soil can inhibit tree growth by dissolving aluminum ions, which damage root hairs. Additionally, aluminum ions entering lakes can kill fish and destroy fish eggs. The diagram captures the overall flow from pollution sources to environmental impacts.

[Generated by AI]

It is easy to observe roughened surfaces and loss of detail in carving in buildings and statues as a result of acid rain. **Figure 7** takes a closer look at the effects of acid rain on a statue.



Figure 7. Effects of acid rain on statues.

Credit: RapidEye, [Getty Images](https://www.gettyimages.co.uk/detail/photo/sad-young-woman-mourning-on-a-1932-grave-monument-royalty-free-image/1159180665)

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International Mindedness

There have been many global efforts to reduce the production of nitrogen and sulfur containing gases. Many industrial facilities reduce sulfur oxides by adding CaCO_3 during the process to produce CaSO_4 either before or after the combustion process. This drastically reduces the sulfur oxides in the atmosphere. The combustion engines of modern vehicles all contain catalytic converters. These aim to return nitrogen containing oxides back to inert nitrogen. The addition of catalytic converters to vehicle combustion engines have had a dramatic impact on the nitrogen oxides in the atmosphere.

Acid rain was a considerable global environmental concern in the 1970s and 1980s and, through collaborative policy making between many nations, effects have been significantly reduced. While still an environmental concern, it is not the urgent environmental crisis it once was. How might societies use lessons learned from acid rain to tackle urgent pressing matters in today's climate fight?

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Designing procedures
- **Time required to complete activity:** 45 minutes
- **Activity type:** Individual activity

Part A: Design lab

You have two mystery elements, one metal and one non-metal. Below is a list of lab equipment you have access to as well as two videos showing the formation of oxides from magnesium and sulfur. Using this information, design an experiment to determine which element is the metal and which is the non-metal. Ensure to carefully plan safety considerations and waste disposal.

- Element 1
- Element 2
- pH or litmus paper
- Bunsen burner
- Tongs
- Test tube holder
- Test tube rack
- Pyrex test tube
- Beaker
- Water
- Fume hood
- Safety glasses

Burning Magnesium Ribbon



Video 1. Burning of magnesium in air.

 More information for video 1

Sulphur Dioxide forms an acid | Acids & Bases | Chemis...



Video 2. Burning of sulfur in air.

Part B: Chemical equations

Write chemical equations for the following:

- Formation of magnesium oxide from magnesium.
- Formation of sulfur dioxide from sulfur.

- Reactions showing both magnesium oxide and sulfur dioxide reacting with water.

Part C: Reading through an article

Read through the abstract, introduction, conclusion and **Figure 1** of this open access article  (<https://link.springer.com/content/pdf/10.1007/s13280-019-01244-4.pdf?pdf=button>). Write down your responses to the following questions:

- What period in time was acid rain of most concern?
- Discuss the progress made in the reduction of nitrogen and sulfur oxide emissions.
- Do you think we should still be concerned about acid rain today?
- Discuss how important you think science-policy interactions are to addressing climate change.
- What are some of the limitations of science-policy interactions?